

The Nancy Grace Roman Space Telescope (shortened as the Roman Space Telescope, Roman, or RST) is a NASA infrared space telescope whose construction was completed on 25 November 2025, and is scheduled to be launched to a Sun–Earth L2 orbit on 30 August 2026. It is named after former NASA Chief of Astronomy Nancy Grace Roman.

The Roman Space Telescope is based on an existing 2.4 m (7.9 ft) wide field of view primary mirror donated by the National Reconnaissance Office and will carry two scientific instruments. The Wide-Field Instrument (WFI) is a 300.8-megapixel multi-band visible and near-infrared camera, providing a sharpness of images comparable to that achieved by the Hubble Space Telescope over a 0.28 square degree field of view, 100 times larger than imaging cameras on the Hubble. The Coronagraph Instrument (CGI) is a high-contrast, small field of view camera and spectrometer covering visible and near-infrared wavelengths using novel starlight-suppression technology.

Stated objectives include a search for extra-solar planets using gravitational microlensing, along with probing the chronology of the universe and growth of cosmic structure, with the end goal of measuring the effects of dark energy, the consistency of general relativity, and the curvature of spacetime.

Roman was recommended in 2010 by the United States National Research Council Decadal Survey committee as the top priority for the next decade of astronomy. It was approved for development and launch on 17 February 2016.

Development of mission

The design of the Roman Space Telescope shares a heritage with various proposed designs for the Joint Dark Energy Mission (JDEM) between NASA and the Department of Energy (DOE).

The original design, called WFIRST (Wide-Field Infrared Survey Telescope) Design Reference Mission 1, was studied in 2011–2012, featuring a 1.3 m (4.3 ft) diameter unobstructed three-mirror anastigmat telescope. It contained a single instrument, a visible to near-infrared

imager/slitless prism spectrometer.

In 2012, another possibility emerged: NASA could use a second-hand National Reconnaissance Office (NRO) telescope made by Harris Corporation to accomplish a mission like the one planned for WFIRST. NRO offered to donate two telescopes, the same size as the Hubble Space Telescope but with a shorter focal length and hence a wider field of view. This provided important political momentum to the project, even though the telescope represents only a modest fraction of the cost of the mission and the boundary conditions from the NRO design may push the total cost over that of a fresh design. This mission concept, called WFIRST-AFTA (Astrophysics Focused Telescope Assets), was matured by a scientific and technical team; this mission is now the only present NASA plan for the use of the NRO telescopes. The Roman baseline design includes a coronagraph to enable the direct imaging of exoplanets.

Several implementations of WFIRST/Roman were studied. These included the Joint Dark Energy Mission-Omega configuration, an Interim Design Reference Mission featuring a 1.3 m (4.3 ft) telescope, Design Reference Mission 1 with a 1.3 m telescope, Design Reference Mission 2 with a 1.1 m (3.6 ft) telescope, and several iterations of the AFTA 2.4 m (7.9 ft) configuration.

In the 2015 final report, Roman was considered for both geosynchronous orbit and for an orbit around the Sun-Earth Lagrange point L2. L2 has disadvantages versus geosynchronous orbit in available data rate and propellant required, but advantages for improved observing constraints, better thermal stability, and more benign radiation environment. Some science cases (such as exoplanet microlensing parallax) are improved at L2, but the possibility of robotic servicing at either of the locations is currently unknown. By February 2016, it had been decided to use a halo orbit around L2.

The project is led by a team at NASA's Goddard Space Flight Center in Greenbelt, Maryland. On 30 November 2018, NASA announced it had awarded a contract for the telescope. This was for a part called the Optical Telescope Assembly (OTA), and ran to 2025. This was in conjunction

with the Goddard Space Flight Center, for which the OTA was planned for delivery as part of the contract.

A February 2019 description of the mission's capabilities is available in a white paper issued by members of the Roman team.

During the 2025 federal government shutdown, NASA's Goddard Space Flight Center began closing multiple buildings and consolidating equipment, which the agency describes as a pre-planned "strategic consolidation" not expected to impact missions. A NASA spokesperson told CNN that work on the Nancy Grace Roman Space Telescope and the Dragonfly mission would continue during the shutdown as excepted operations.

According to Space.com, some employees at the center expressed concern that the manner and timing of the closures could disrupt work on those projects, and CNN reported that an internal document compiled by Goddard staff cited potential impacts on specialized testing laboratories.

Science objectives

The science objectives of Roman aim to address cutting-edge questions in cosmology and exoplanet research, including:

Answering basic questions about dark energy, complementary to the European Space Agency (ESA) Euclid mission, and including: Is cosmic acceleration caused by a new energy component or by the breakdown of general relativity on cosmological scales? If the cause is a new energy component, is its energy density constant in space and time, or has it evolved over the history of the universe? Roman will use three independent techniques to probe dark energy: baryon acoustic oscillations, observations of distant supernovae, and weak gravitational lensing.

Completing a census of exoplanets to help answer new questions about the potential for life in the universe: How common are solar systems like our own? What kinds of planets exist in

the cold, outer regions of planetary systems? What determines the habitability of Earth-like worlds? This census makes use of a technique that can find exoplanets down to a mass only a few times that of the Moon: gravitational microlensing. The census would also include a sample of free-floating planets with masses likely down to the mass of Mars.

Establishing a guest investigator mode, enabling survey investigations to answer diverse questions about our galaxy and the universe.

Providing a coronagraph for exoplanet direct imaging that will provide the first direct images and spectra of planets around our nearest neighbors, similar to our own giant planets.

Detection of primordial black holes.

Instruments

The telescope is to carry two instruments. Both were installed into the telescope in December 2024.

WFI

The Wide-Field Instrument (WFI) is the primary science instrument for the mission. It is a 300.8-megapixel camera, providing multiband imaging and spectroscopy from the visible to near-infrared (0.48 to 2.30 μm). Eight science filters, one wide band, and seven more narrow filters, provide high-quality images in subsets of the total spectral bandwidth of the instrument. It also carries both a high-dispersion grism and lower-dispersion prism assemblies for wide-field slitless spectroscopy. A "dark element" made out of spectralon can block out the light from space and reflect calibration light onto the detector. The eight filters, two spectroscopy elements, and dark element are all attached to an Element Wheel Assembly (EWA) that rotates one element at a time into the active position. This enables the scientists to choose which colors of light are needed for each mission, or switch between imaging and

spectroscopy modes. A HgCdTe-based focal-plane array (FPA) captures a 0.28 square degree field of view with a resolution of 0.11 arcseconds. The detector array is composed of 18 H4RG-10 detectors provided by Teledyne. The instrument carries on-board calibration capability to help the scientists discern the subtle changes in images from their dark matter mission compared to ordinary distortions of the optics or of the FPA. The FPA is attached to a set of hexapod actuators that can reposition the focal plane into the best focus position based on analysis from the ground to ensure the captured images are always of high quality over the lifetime of the mission. WFI was completed and shipped to NASA in August 2024.

CGI

The Coronagraph Instrument (CGI) is a high contrast coronagraph covering shorter wavelengths (575 nm to 825 nm) using dual deformable mirror starlight-suppression technology. It is intended to achieve a part-per-billion suppression of starlight to enable the detection and spectroscopy of planets with a visual separation of as little as 0.15 arcseconds from their host stars. CGI is intended as a technology demonstrator for an exoplanet imaging instrument on a future large space-based observatory, such as the Habitable Worlds Observatory (HWO).

History

On 2 March 2020, NASA announced that it had approved WFIRST to proceed to implementation, with an expected development cost of US\$3.2 billion and a maximum total cost of US\$3.934 billion, including the coronagraph and five years of mission science operations.

On 20 May 2020, NASA Administrator Jim Bridenstine announced that the mission would be named the Nancy Grace Roman Space Telescope in recognition of the former NASA Chief of Astronomy's role in the field of astronomy.

On 31 March 2021, the NASA Office of Inspector General (OIG) released a report that stated

that the development of the Nancy Grace Roman Space Telescope had been affected by the COVID-19 pandemic, which hit the US during a particularly important time in the telescope's development. NASA is expecting a total impact of US\$400 million due to the pandemic and its effect on subcontractors for the project.

On 29 September 2021, NASA announced that Roman had passed its Critical Design Review (CDR), and that with predicted impacts from COVID-19 disruptions, and with flight hardware fabrication completed by 2024 followed by mission integration, the launch date would be no later than May 2027.

On 19 July 2022, NASA announced that Roman would be launched on a Falcon Heavy launch vehicle, with a contract specifying readiness by October 2026 and a launch cost of approximately \$255 million.

In September 2024, the satellite bus which will carry the telescope, was substantially completed. The following December, the instruments and mirror assembly were successfully integrated onto a section called the "instrument carrier".

In October 2024, the telescope passed a major 'spin test'. The space telescope had finished construction on 25 November 2025.

On 21 June 2026, the completed telescope arrived at the Kennedy Space Center, in preparation for launch.

Funding history and status

In the fiscal year 2014, Congress provided US\$56 million for Roman, and in 2015 Congress provided US\$50 million. The fiscal year 2016 spending bill provided US\$90 million for Roman, far above NASA's request of US\$14 million, allowing the mission to enter the "formulation phase" in February 2016. On 18 February 2016, NASA announced that Roman had formally become a project (as opposed to a study), meaning that the agency intends to carry out the mission as baselined; at that time, the "AFTA" portion of the mission concept name

WFIRST-AFTA was dropped, as only that approach was being pursued with a plan for a mid-2020s launch. The total cost of Roman at that point was expected at more than US\$2 billion; NASA's 2015 budget estimate was around US\$2.0 billion in 2010 dollars, which corresponds to around US\$2.7 billion in real year (inflation-adjusted) dollars.

The mission began Phase A development in 2016 with Neil Gehrels, as project scientist, chairing the Formulation Science Working Group with co-chairs Jeremy Kasdin and David Spergel. In April 2017, NASA commissioned an independent review of the project to ensure that the mission scope and cost were understood and aligned. The review acknowledged that Roman offers "groundbreaking and unprecedented survey capabilities for dark energy, exoplanet, and general astrophysics", but directed the mission to "reduce cost and complexity sufficient to have a cost estimate consistent with the US\$3.2 billion cost target set at the beginning of Phase B". NASA announced (January 2018) the reductions taken in response to this recommendation, and that Roman would proceed to its mission design review in February 2018 and begin Phase B by April 2018. NASA confirmed (March 2018) that the changes made to the project had reduced its estimated life cycle cost to US\$3.2 billion and that the Phase B decision was on track to begin on 11 April 2018.

In February 2018, the Trump administration proposed an FY2019 budget that would have delayed the funding of the Roman (then called WFIRST), citing higher priorities within NASA and the increasing cost of the telescope. The proposed defunding of the project was met with criticism by professional astronomers, who noted that the American astronomical community had rated Roman the highest-priority space mission for the 2020s in the 2010 Decadal Survey. The American Astronomical Society expressed "grave concern" about the proposed defunding, and noted that the estimated lifecycle cost for Roman had not changed over the previous two years. In agreement, the United States Congress approved an FY2018 Roman budget on 22 and 23 March 2018 in excess of the administration's budget request for that year, stating that it "rejects the cancellation of scientific priorities recommended by the National Academy of

Sciences decadal survey process", and directed NASA to develop new estimates of Roman's total and annual development costs. The President of the United States announced he had signed the bill on 23 March 2018. NASA was funded via a FY2019 appropriations bill on 15 February 2019, with US\$312 million for Roman, rejecting the President's reduced Budget Request and reasserting the desire for completion of Roman with a planning budget of US\$3.2 billion.

In March 2019 the Trump administration again proposed to defund the Roman in its FY2020 budget proposal to Congress. In testimony on 27 March 2019, NASA Administrator Jim Bridenstine hinted that NASA would continue Roman after the James Webb Space Telescope, stating "WFIRST will be a critical mission when James Webb is on orbit". In a 26 March 2019, presentation to the National Academies' Committee on Astronomy and Astrophysics, NASA Astrophysics Division Director Paul L. Hertz stated that Roman "is maintaining its US\$3.2 billion cost for now ... We need US\$542 million in FY2020 to stay on track". At that time, it was stated that Roman would hold its Preliminary Design Review (PDR) for the overall mission in October 2019 followed by a formal mission confirmation in early 2020.

NASA announced the completion of the Preliminary Design Review (PDR) on 1 November 2019, but warned that though the mission remained on track for a 2025 launch date, shortfalls in the Senate's FY2020 budget proposal for Roman threatened to delay it further.

In April 2025, the second Trump administration proposed to cut funding for Roman again as part of its FY2026 budget draft. This was part of wider proposed cuts to NASA's science budget, down to US\$3.9 billion from its FY2025 budget of US\$7.5 billion. On 25 April 2025, the White House Office of Management and Budget announced a plan to cancel dozens of space missions, including the Roman Space Telescope, as part of the cuts. These cuts are still being debated in the United States Congress.

Institutions, partnerships, and contracts

The Roman project office is located at NASA's Goddard Space Flight Center (GSFC) in Greenbelt, Maryland, and holds responsibility for overall project management. GSFC also leads the development of the Wide-Field Instrument, the spacecraft, and the telescope. The Coronagraph Instrument is being developed at NASA's Jet Propulsion Laboratory in Pasadena, California. Science support activities for Roman are shared among the Space Telescope Science Institute in Baltimore, Maryland, which is the Science Operations Center; the Infrared Processing and Analysis Center in Pasadena, California; and GSFC.

Partners

Four international partners, namely the French space agency CNES, European Space Agency (ESA), Japan Aerospace Exploration Agency (JAXA), and the Max Planck Institute for Astronomy have joined with NASA to provide various components and science support for Roman. Beginning in 2016 NASA expressed interest in ESA contributions to the spacecraft, coronagraph and ground station support. For the coronagraph instrument, contributions from Europe and Japan have been established. In 2018, a contribution from Germany's Max Planck Institute for Astronomy (MPIA) was under consideration, namely the filter wheels for the star-blocking mask inside the coronagraph. In 2016, the Japanese space agency JAXA proposed to add a polarization module for the coronagraph, plus a polarization compensator. An accurate polarimetry capability on Roman may strengthen the science case for exoplanets and planetary disks, which shows polarization. Ground support will be provided by a new NASA station in White Sands in New Mexico, the Misasa station in Japan and ESA's New Norcia station in Australia.

Construction contracts

In May 2018, NASA awarded a multi-year contract to Ball Aerospace (now called BAE Systems Space & Mission Systems) to provide key components (the Opto-Mechanical Assembly) for the

Wide-Field Instrument on Roman. In June 2018, NASA awarded a contract to Teledyne Scientific and Imaging to provide the infrared detectors for the Wide-Field Instrument. On 30 November 2018, NASA announced it had awarded the contract for Optical Telescope Assembly to the Harris Corporation of Rochester, New York. Via an agreement signed in June 2022, MPIA designed, built, and provided all of CGI's Precision Alignment Mechanisms (PAM) which operate, stabilize, and control optical elements such as optical filters.

Gallery

See also

Wide-field Infrared Survey Explorer – NASA satellite of the Explorer program

Spitzer Space Telescope – NASA infrared space telescope (2003–2020)

Xuntian – Planned Chinese space telescope

References

External links

Roman Science Community page at Goddard Space Flight Center site

Roman Science Data Center page at the Infrared Processing and Analysis Center (IPAC)

US\$1.6 Billion Telescope Would Search Alien Planets and Probe Dark Energy – Space.com

The WFIRST/AFTA astrophysics mission: bigger and better for exoplanets Archived 5 April 2016 at the Wayback Machine, Tom Greene

"Astro2010 Report Release Presentation". Archived from the original on 26 July 2011. Retrieved 15 August 2010.

NASA/ Goddard – WFIRST: Uncovering the Mysteries of the Universe on YouTube (min. 1:25) 30 May 2014

WFIRST-AFTA: Coronagraph Technology Development on YouTube (min. 4:20) 16 March 2015

WFIRST: The Best of Both Worlds on YouTube (min. 3:14) 18 February 2016

Introducing NASA's Roman Space Telescope on YouTube (min. 1:28) 25 March 2026